0625 | PAPER 4

TOPIC 06
Space Physics

27 topical answers • 2021–2025
Calculations, final values and examiner-keyword summaries

Answers are independently condensed from the official marking requirements. They are not a reproduced Cambridge mark scheme. For drawings and graph lines, compare the stated method with the original question figure.

Topic contents

Core formulas

- orbital speed = $2\pi r \div$ period
- Hubble relation: $v = H_0 d$
- redshift shows distant galaxies are receding

6.1 Solar System and accretion model

Q327 • Feb/March 2023 Paper 42 Question 10
6.1 Solar System and accretion model

(a) 24 km / s

v = 2πr / T or (v =) 2πr / T or (2π × 2.28 × 108) / (690 × 24 × 60 × 60)

(2π × 2.28 × 108) / (690 × 24 × 60 × 60) or (T =) 690 × 24 × 60 × 60 or (T=) 59 616 000 (s)

(b) elliptical / ellipse

(c)(i) wavelength (of light from distant galaxies) increases

occurs when galaxies are moving away (from Earth)

(c)(ii) speed / velocity (that galaxy is moving away from Earth)

Q328 • May/June 2023 Paper 41 Question 10
6.1 Solar System and accretion model

(a)(i) (speed) decreases (from X to Y) and then increases (from Y to X)

(a)(ii) Any three valid point(s):

• gravitational (potential) energy (GPE) transfers to kinetic energy (KE) or vice versa
• KE transfers to GPE from X to Y and GPE transfers to KE from Y to X
• speed decreases as KE decreases /
• most GPE at Y or least GPE at X
• total (of GPE + KE) energy is constant

(b)(i) -230 (°C)

(b)(ii) (white surface) is a poor absorber / good reflector / poor emitter of IR / radiation

or black / other surface is a good absorber / poor reflector / good emitter of IR / radiation

Any one valid point(s):

• (the white surface) increases in temperature less when facing the Sun
• (the white surface) decreases in temperature less when facing away (from Sun)
• the black / other surfaces increases in temperature more when facing the Sun
• the black / other surface decreases in temperature more when facing away (from Sun)
• less variation in temperature on white surface (during one whole rotation)

Q329 • May/June 2023 Paper 42 Question 10
6.1 Solar System and accretion model

(a) v = 2πr / T

r = (average) radius of the orbit and T = (orbital) period

(b) rays from Sun strike the country at different angles through the year

rays from Sun strike the country for different number of hours per day through the year

(c) (first space:) red supergiant

(second space:) nebula

(3rd and 4th spaces:) neutron star

black hole

(d) 1.6 × 109 (light-years)

H0 = v / d or (d =) v / H0 or (d =) [33 000 × 103] / [2.2 × 10-18 × 9.5 × 1015]

Q330 • May/June 2023 Paper 43 Question 9
6.1 Solar System and accretion model

(a) both positions correctly marked S1 and J1 on the diagram

Saturn moved 360 × 5 / 30 (= 60°) or S1 in correct position

Jupiter moved 360 × 5 / 12 (= 150°) or J1 in correct position

(b)(i) S2 at 240° and J2 at (600° - 360° =) 240°

(b)(ii) (Saturn and Jupiter) are aligned / Jupiter exactly in front of Saturn / there is a conjunction

(c)(i) (Jupiter) gaseous and large

(Earth) rocky and small

(c)(ii) Any three valid point(s):

(density)

• Key physics: Jupiter has low density because composed gas / Earth high solid

(gravitational field strength)
• Key physics: Jupiter has large GFS so mass / Earth small
• Jupiter's mass is larger than the Earth's mass because the volume of Jupiter is larger even though the density of Jupiter is smaller

(d) (mass =) 1.8 × 1027 kg

ρ = m / V or (m =) ρV or (m =) 1300 × 1.4 × 1015 × 109
(m =) 1300 × 1.4 × 1015 × 109 (kg) or 1.8(2) × 10N

Q331 • Oct/Nov 2023 Paper 42 Question 9 — 6.1 Solar System and accretion model

(a) Venus
(b) The larger the mass (of the planet), the larger the gravitational field strength (at the surface)
(c) orbit of planets is elliptical / is not circular
(d) correct conversion of T into seconds i.e. 365.2 × (24 × 60 × 60) or 3.2 × 107

(v =) {2πr} / T
2π × 149.6 × 106 / 365.2 × 24 × 60 × 60

Q332 • May/June 2024 Paper 42 Question 10 — 6.1 Solar System and accretion model

(a) Any two valid point(s):
• minor planets or dwarf planets
• comets
• asteroids

(b) elliptical
(c) kinetic energy (store) decreases and potential energy (store) increases (as object moves from A to B)

energy is conserved

(d) 2.6 × 103 s

v = s / t or (t =) s / v or 7.8 × 1011 / 3.0 × 108

Q333 • Oct/Nov 2024 Paper 41 Question 9 — 6.1 Solar System and accretion model

(a) group / collection of (billions of) stars
(b) 9.5 × 1017 (km)

(1 light-year =) 9.5 × 1015 (m) or (1 light-year =) 3 × 108 × 365 × 24 × 3600

(c)(i) increase in wavelength (of light from far galaxy) or (amount of) redshift
(c)(ii) brightness of a supernova
(c)(iii) (their) speeds are (directly) proportional to distances (from Earth) or H0= v / d
(c)(iv) 4.5 × 1017 (s)

(age of Universe =) 1 / H0 or 1 / (2.2 × 10-18)

Q334 • Oct/Nov 2024 Paper 42 Question 11 — 6.1 Solar System and accretion model

(a)(i) Earth / Mars / Jupiter / Saturn / Uranus / Neptune
(a)(ii) Mercury
(b) 3.6 × 104 (m / s)

(T =) 220 × 24 × 60 × 60 or (T =) 1.9 × 107 (s)
2πr 2π × 1.1× 1011
(v =) or (v =)
T 220 × 24 × 60 × 60

(c) The further away from the Sun the slower the orbital speed /
(d) 1 Any one valid point(s):

• comet has an elliptical orbit
• speed of comet is faster when it closer to the Sun
• speed of comet is slower when it is further away from the Sun
2 Any one valid point(s):
• (conservation of energy requires that) transfers between kinetic and gravitational stores (as comet changes speed)

- total energy remains constant
- energy cannot be created or destroyed

3 as radius of orbit decreases, gravitational energy decreases and kinetic energy increases

Q335 • Feb/March 2025 Paper 42 Question 10

6.1 Solar System and accretion model

(a) (number of days =) 91

(statement F to G) is ¼ of (a complete) orbit or a whole year is 365 days

(b)(i) F

(b)(ii) H

(c) 1.5 × 1011 m

v = 2πr/T or (r =) vT/(2π) or (r =) 3.0×104 ×365 × 24 × 60 × 60 / 2π

(T =) 365 × 24 × 60 × 60 or (T =) 31 536 000 or (r =) 3.0 × 104 × T / (2π) or

correct rearrangement of and substitution into formula using candidate's value of T

(d) Any one valid point(s):

- minor planets
- asteroids or meteoroids
- moons (that orbit planets)
- comets
- natural satellites

Q336 • May/June 2025 Paper 41 Question 10

6.1 Solar System and accretion model

(a) Jupiter is gaseous

Earth is rocky

(b)(i) weight

(gravitational) force per unit mass or (g =) in this form

mass

(gravitational) force on a mass or W = mg

(b)(ii) mass

(c) (orbital speed of) Jupiter is slower

Jupiter is further from the Sun; or orbital speeds of planets decrease as distance from the Sun increases

gravitational field (strength) of Sun decreases with distance (from Sun)

Q337 • May/June 2025 Paper 42 Question 10

6.1 Solar System and accretion model

(a)(i) planet B

high(est) (surface) temperature

(a)(ii) -180 (°C)

(a)(iii) planet A

gravitational field strength is highest so acceleration (due to gravity) is greater

gravitational field strength is highest so greater (downward) force on the object

(b)(i) red supergiant

(new) heavier elements or hydrogen

Any two valid point(s):

- neutron star
- black hole
- new stars (with orbiting planets)

(b)(ii) distance (from Earth) to galaxy

Q338 • Oct/Nov 2025 Paper 42 Question 10

6.1 Solar System and accretion model

(a) rocky

gaseous

gravitational force

accretion disc

(b) 3.9 × 108 (m)
(speed of light =) 3.0 × 108
(s =) vt or (s =) 3.0 × 108 × 1.3
(c) 4.8 × 105 m / s or 4.8 × 102 km / s
(1 light-year =) 9.5 × 1015 (m) or 9.5 × 1012 (km) or 4.8 × 10N
(v =) H0 d or (v =) 2.2 × 10-18 × 2.3 × 107 × 9.5 × 1015

Q339 • Oct/Nov 2025 Paper 43 Question 9
6.1 Solar System and accretion model

(a)(i) (shortly after) Big Bang or when Universe was formed
(a)(ii) Increased
Universe has expanded
Universe accelerated outwards
(b)(i) 120 000 (km / h) or 1.2 × 105 (km / h)
(v =) 2πr / T or {2 × π × 108 (× 106) } / {0.62 × 365 × 24}
(T =) 0.62 × 365 × 24 or (T =) 5431 or (T =) 226.3 × 24 or 1.2 × 10n
(b)(ii) decreases

6.2 Comets and orbits

Q340 • Oct/Nov 2025 Paper 43 Question 10
6.2 Comets and orbits

(a)
S or S
elliptical shape drawn
S at a focus of the ellipse by eye
(b)
S L
L furthest away from their S / Sun
(c) total energy = gravitational (potential) energy + kinetic energy
(energy stored as) GPE decreases and (energy stored as) KE increases closer to the Sun
when the comet has more kinetic energy it has a greater speed

6.3 Stars, galaxies and the Universe

Q341 • May/June 2023 Paper 43 Question 10
6.3 Stars, galaxies and the Universe

(a) v = s / t or (s =) vt
(c =) 3 × 108 (m / s)
(1 year =) 365 × 24 × 3600 (s) or 3.2 × 107 (s) or 32 × 106 or 8760 × 3600
(s =) candidate's speed of light × candidate's time (m)
(b)(i) change of wavelength (of galaxy's starlight) or redshift
(b)(ii) H0 = v / d
(b)(iii) brightness of a supernova

Q342 • Oct/Nov 2023 Paper 41 Question 10
6.3 Stars, galaxies and the Universe

(a) hydrogen nuclei fuse to become helium nuclei
nuclear reactions or (nuclear) fusion
hydrogen fuses into helium
(b)(i) (observed) wavelength is longer / wavelength is shifted towards the red end of the spectrum
(light from galaxy) redshifted / shifted towards red (end of spectrum)
(b)(ii) change in wavelength (or starlight due to redshift)
(c)(i) 5.9 × 1024 m

H0 = v/d or (d =) v / H0 or 1.3 × 107 / 2.2 × 10-18 or 5.9 × 10N (m)

(c)(ii) 1.4 × 1010 (years)

(age =) 1 / H0 or 1 / 2.2 × 10-18 or 4.5 × 1017

Q343 • Oct/Nov 2023 Paper 42 Question 10
6.3 Stars, galaxies and the Universe

(a) (interstellar clouds of) gas and dust or (stellar) nebula

(b) (inward) force of gravitational attraction (is balanced by)

(outward) force due to the high temperature (in the centre of the star)

(c) hydrogen

(d) planetary nebula

Q344 • Oct/Nov 2023 Paper 43 Question 10
6.3 Stars, galaxies and the Universe

(a)(i) 1 Hubble constant

2 H0 = v / d

(a)(ii) per second or s-1 or 1 / s

(a)(iii) 4.5 × 1017 (s)

d / v = 1 / H0

or (age of Universe =) 1 / H0

or (age of Universe =) d / v

or (age of Universe =) 1 / 2.2 × 10--18

(b) shortly after the Universe was formed

or shortly after the Big Bang

all points in space

Q345 • May/June 2024 Paper 41 Question 9
6.3 Stars, galaxies and the Universe

(a) ultraviolet and visible light and infrared only

Any two valid point(s): ultraviolet; visible light; infrared and no more than one incorrect addition

(b) 1.5 × 1011 m

v = s / t or (s =) vt or 3.0 × 108 × 500 or 1.5 × 10N

(c)(i) Any two valid point(s):

- cloud / nebula / it collapses
- due to (internal) gravitational attraction
- (internal) temperature increases

(c)(ii) Any three valid point(s):

- (nuclear) fusion / nuclear reactions (in the star)
- forces are balanced
- gravitational force is inwards
- outwards force is due to high temperature

Q346 • May/June 2024 Paper 42 Question 11
6.3 Stars, galaxies and the Universe

(a) Milky Way

(b) Big Bang (Theory)

(c)(i) shortly after the Universe was formed

(c)(ii) Universe has expanded

(radiation) has been redshifted (to the microwave region of the electromagnetic spectrum)

Q347 • May/June 2024 Paper 43 Question 11
6.3 Stars, galaxies and the Universe

(a) redshift

(b)(i) (H0 =) 2.2 × 10-18 per second

(b)(ii) 2.6 × 108 m / s

H0 = v / d or (v =) H0d or (v =) 1.2 × 1026 × 2.2 × 10-18

(b)(iii) Any two valid point(s):

- more distant galaxies are moving away faster
- universe is expanding
- (if galaxies are moving away from each other now, then) in the past galaxies were closer together

Q348 • Oct/Nov 2024 Paper 43 Question 10 6.3 Stars, galaxies and the Universe

(a) Any two valid point(s):

- (most of the) hydrogen has been converted to helium or hydrogen begins to run out
- (star expands and) forms a red supergiant
- (red supergiant) explodes as a supernova (forming a nebula)

core (of red supergiant collapses and) forms black hole; or a black hole forms at the centre (of the supernova/nebula)

(b)(i) the distance travelled (in the vacuum of space) by light in one year

(b)(ii) 7.4 × 1016 km

9.5 × 1015 (m) or 9.5 × 1012 (km)

Q349 • May/June 2025 Paper 43 Question 10 6.3 Stars, galaxies and the Universe

(a) distance travelled (in the vacuum of space) by light in one year

(b) 1.5 × 10 1 1 m

9.5 × 1015 (m) or (c =) 3(.0) × 108 (m / s) or 3(.0) × 10N × 490

(c) 2.5 × 10-1 8 per second

(Hubble's constant =) gradient or (H0 =) v ÷ d

2.5 × 104(- 0) ÷ {100 × 1020 (- 0)}

(d) electromagnetic radiation / light from (distant) galaxies

(observed) increase in wavelength (compared to wavelength measured on the Earth)

6.4 Space physics - mixed

Q350 • Oct/Nov 2025 Paper 41 Question 10 6.4 Space physics - mixed

(a) B

Sun is not at the centre of the orbit or comets have an elliptical orbit

(b) Comet travels faster when it is closer to the Sun

GPE decreases and KE increases as comet gets closer to the Sun

(c)(i) distance travelled (in the vacuum of space) by light in one year.

(c)(ii) 6.3 × 1010 (m) or 63 000 000 000 (m)

(1 light-year =) 9.5 × 1015 (m) or (1 light-year =) 9.5 × 1012 (km) or SEEN (distance =) 6.6 × 10-6 × 9.5 × 10N

6.5 Moon phases and orbits

Q351 • Feb/March 2024 Paper 42 Question 10 6.5 Moon phases and orbits

(a)(i) position A and / or E

position G

(a)(ii) 1 month

(b)(i) 110 000 (km / h)

(v =)2πr / T

T = 365 × 24h or 2π × 1.5 × 108 / 365 × 24

(b)(ii) 500 s

v = s / t or (t =) s / v or 1.5 × 1011 / 3.0 × 108

6.6 Planets and the Solar System

Q352 • Oct/Nov 2025 Paper 41 Question 2 — 6.6 Planets and the Solar System

(a) (gravitational field strength is) force per unit mass (on an object in a gravitational field)

(b)(i) 16 N

W = mg or (m =) W÷g or 42 / 9.8 or (mass of object =) 4.3 (kg)

(b)(ii) 1.6 × 1020 m3

(V =) m / ρ or (V =) 6.4 × 1023 / 3900

(c)(i) arrow parallel to driving force and pointing to the right

(c)(i) (arrow pointing to the right) labelled 30 N

(c)(ii) 1 (resultant force) increases or there is a resultant force (in the direction of the driving force)

resultant force = driving force - resistive force(s)

6.7 Stars and the Universe

Q353 • Feb/March 2024 Paper 42 Question 11 — 6.7 Stars and the Universe

(a) inward force / force of gravitational attraction, is balanced by an outward force / force due to fusion reactions

(b)(i) ratio of the speed at which the galaxy is moving away from the Earth/observer to its distance (from the Earth/observer)

ratio of speed (of a galaxy) to distance (away from observer)

(b)(ii) (age of universe =) d / v = 1 / H0 or age (of universe) = 1 / H0

(b)(iii) 4.5 × 1017 (s)